

Fran Menéndez Moya

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Zaragoza, Spain | EU work authorization | Remote

SUMMARY

Software Engineer with 10+ years building platforms that serve millions of users. I work across the full stack and increasingly at the AI layer: semantic search in production, MCP tooling, and agentic engineering workflows I have made team-wide standards. I lead architecture domain-wide, mentor engineers, and set technical standards that scale.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, HTML/CSS, SQL
AI Engineering: Semantic Search (OpenSearch k-NN, vector embeddings, BM25), MCP (tooling & design-to-code), Agentic Development Workflows (multi-agent orchestration, model routing, adversarial AI code review), Context Engineering
Application Security: SQL injection prevention, access control & IDOR, PII exposure, security regression testing in CI
Frontend: Next.js, React, Vue.js, Angular, Astro, PWAs
Backend: Node.js, NestJS, Microservices, Real-time Systems, REST, GraphQL
Data & Search: PostgreSQL, MongoDB, OpenSearch, Redis
Infrastructure: AWS (Cognito, Lambda, CloudFront, S3), Kubernetes, Docker, CDN, Message Queues
CMS & Commerce: PayloadCMS, Contentful (incl. custom editor apps), Headless Architecture
CI/CD & Tools: Jenkins, GitHub Actions, Git, Automated Pipelines

PROFESSIONAL EXPERIENCE

The Knot Worldwide Remote | Oct 2023 – Present

Principal Software Engineer Apr 2025 – Present

- Rebuilt the Shop e-commerce platform from scratch in one month solo (versus a three-month, multi-developer estimate). The platform handles millions of products from Amazon's Rainforest API and editorial content via Contentful, built with Next.js and React with shared component libraries, now serving hundreds of thousands of users in production.
- Built a semantic product matching system using OpenSearch with k-NN vector similarity and BM25 text relevance, with a custom human-review dashboard inside PayloadCMS for matches below confidence thresholds.
- Built the team's agentic AI development workflow: orchestration and verification architecture with role-based model routing, context budgeting, and mandatory independent review; used for production delivery including security-critical work.
- Introduced Model Context Protocol (MCP) tooling that generates production components directly from Figma at token-exact fidelity, removing the manual design-to-code translation step; now the team-wide standard.
- Ran a full API security audit and hardening programme for the public e-commerce service: eight findings across four vulnerability classes, closing SQL injection, broken access control, a write-side IDOR and a PII exposure; built the service's first automated test harness (two-tier, DB-free CI proof layer) and a regenerable security-regression tripwire in CI.
- Built a custom Contentful editor app for the editorial platform (React, Contentful App SDK), including its S3 + CloudFront hosting infrastructure; propagated the new content model end-to-end across three services with zero-downtime migrations.
- Migrated PayloadCMS from v2 to v3 for 100,000+ products with zero downtime; query response times dropped from 850ms to 34ms through database optimization and caching architecture.
- Reworked caching for 2M+ weekly users: added a cache layer and set per-endpoint revalidation and staleness policies after investigating each endpoint's tolerance, on the existing CDN with no added infrastructure; cut cost with no availability regression.
- Steadied the engineering team through a company restructuring, and argued build-capacity against target-attainment in the half-year OKR capacity review; adjudicated review disputes on evidence rather than seniority.
- Root-caused cross-service authentication failures across the platform and shop boundary, isolating an edge-layer fault and a hanging Cognito session call; resolved a Kubernetes-served canonical SEO incident.

Lead Software Engineer Oct 2023 – Mar 2025

- Built an A/B testing system for a platform where edge caching had made experimentation impossible: Akamai assigns a variant cookie at the edge, the app reads it to render the matching tagged build, and assignment stays cached and sticky across reloads until the cookie expires.

- Cut deployment time from over 2 hours to 1 minute, enabling daily deployments and dramatically reducing developer cycle time.
- Led development of the integrated commerce platform, taking it from proposal to a live MVP.
- Managed an 8-person cross-functional team; mentored 5 developers, resulting in 2 promotions.

MOBIKO GmbH

Remote | Aug 2020 – Sep 2023

Team Lead & Architecture Mar 2023 – Sep 2023

- Set the technical strategy and built the microservices architecture behind the company's enterprise expansion, processing 500k+ daily mobility events.
- Grew the engineering team and shortened delivery cycles by restructuring the review and release process and mentoring engineers into ownership.

Senior Full-Stack Developer Aug 2020 – Mar 2023

- Grew the user base several-fold through performance optimization and strategic feature development.
- Cut API response times substantially with caching and database query optimization.
- Automated the deployment pipeline from multi-day release cycles to continuous deployment.

Hiberus Tecnología

Zaragoza, Spain | Jul 2017 – Jul 2020

Senior Full-Stack Developer Jan 2020 – Jul 2020

- Rebuilt the checkout path for a real-time ticketing platform under La Liga match-day load, integrating the official API for 100k+ concurrent users.

Full-Stack Developer Nov 2017 – Jan 2020

- Built a platform processing 1M+ daily entries with real-time sync across 5 geographic locations.

Junior Developer, E-commerce Jul 2017 – Nov 2017

Earlier: Freelance Web Developer 2016 – Jul 2017

Rebuilt the site and built a photo-management admin panel for a university graduation photography studio.

RECOGNITIONS & AWARDS

NASA Space Apps Global Finalist & Galactic Impact Award | Google Hash Code Hub Winner | ImagineCode Blockchain Challenge Winner | 100 Ideas Zaragoza (Most Innovative Technology, Best Use of Citizen Card) | uCode by Adidas Best Fan Experience Award

EDUCATION & LANGUAGES

Computer Engineering (Software Engineering Specialization), [Universidad de Zaragoza](#)

Spanish (Native) | English (Advanced, Cambridge C1)